AI Research & Knowledge Workspace

Comprehensive Technical Report & Verification Assessment

ResearchSpace System Architecture & Verification • November 2026

1. Executive Overview

The AI Research and Knowledge Workspace (ResearchSpace) is an integrated research engineering platform designed to process, analyze, semantically retrieve, and compare academic and enterprise documents. The architecture combines Google Gemini 1.5 generative models, PostgreSQL native tsvector full-text search, deterministic lexical similarity algorithms, and a persistent canvas-based PDF reader.

Key technical capabilities verified in this release:

- **Document Intelligence:** Automated extraction of executive summaries, key takeaways, active-recall study flashcards, quiz questions, and terminology definitions using Gemini 1.5 Pro/Flash.
- **Deterministic Lexical Diff Engine:** Zero-cost comparison of document vocabulary and passages using Jaccard and Cosine Term-Frequency similarity algorithms without LLM hallucinations or credit usage.
- **Persistent In-Browser PDF Reader:** Native canvas rendering with transparent selection text layer, 4-color persistent highlights (yellow, green, blue, pink), attached reflection notes, and rapid Flashcard Revision Mode.
- **Grounded Document & Multi-Doc RAG:** 768-dimensional embeddings generated with text-embedding-004 and queried via PostgreSQL cosine distance with strict page-level source citations.
- **Cryptographic Integrity & Versions:** SHA-256 duplicate detection before physical file creation, alongside document version history with seamless rollback.

2. System Architecture & Stack

The project is organized into three decoupled layers:

- **Frontend (React 18 + Vite):** Responsive user interface with dark and light themes, in-browser PDF canvas viewer, lexical diff panels, and multi-doc chat modal.
- **Backend (Express.js + Node.js):** REST API supporting byte-range requests (HTTP 206 Partial Content), safe config verification, and multer file uploads.
- **AI & Persistence Tier (PostgreSQL + Gemini):** Relational database housing documents, tsvectors, real[] embeddings, and analyses, connected to Gemini 1.5 Flash.

3. Database Schema Specification

All persistent records in ai_knowledge_workspace use UUID primary keys generated by pgcrypto to prevent ID collision and avoid integer enumeration attacks:

Table Name	Key Columns & Types	Functional Purpose
documents	UUID, filename, file_path, file_size, extracted_text, status, search_vector (tsvector)	Core Document Store
document_analyses	UUID, document_id (UNIQUE), summary, key_points, flashcards, quiz_questions	Structured AI Outputs
document_chunks	UUID, document_id, chunk_index, content, page_number, embedding (REAL[768])	RAG Vector Database
document_annotations	UUID, document_id, page_number, selected_text, note_text, color, position	User Highlights & Notes
document_versions	UUID, document_id, version_number, original_filename, file_sha256, created_at	Revision History & Rollback

4. Deterministic Lexical Comparison Engine

Rather than relying on non-deterministic LLM output, ResearchSpace executes mathematical statistical algorithms:

A. Jaccard Vocabulary Similarity:

J(A, B) = |A ") B| / |A "* B|. Evaluates the proportion of shared unique vocabulary tokens between two documents excluding common English stop words.

B. Cosine Term-Frequency Similarity:

Cosine(u, v) = (u · v) / (||u|| * ||v||). Measures thematic alignment across normalized word frequency vectors. High cosine similarity guarantees consistent topic focus.

C. Page-Aligned Passage Diff Alignment:

Sentences are extracted from Document A and Document B. A matching matrix identifies passages with >= 65% token Jaccard similarity as shared (green). Remaining sentences are color-coded as unique to Document A (red) or unique to Document B (blue).

5. Regression Test Suite & Verification Results

A complete 9-stage verification test was conducted against the live PostgreSQL database and backend endpoints (test_regression.js):

Test Suite Stage	Observed Verification Result	Result Status
TEST 1: Database Health	PostgreSQL connected successfully; Gemini API key configured	PASSED (200 OK)
TEST 2: Document Listing	Retrieved uploaded documents (UNIT5.pdf, IOT Notes.pdf)	PASSED (2 docs)
TEST 3: Duplicate Protection	Cached analysis returned immediately; zero redundant Gemini calls	PASSED (Cached)
TEST 4: Single-Doc RAG Chat	4 grounded chunk citations returned with accurate page references	PASSED (Grounded)
TEST 5: Multi-Doc RAG Chat	6 citations cited across both workspace documents	PASSED (Multi-Doc)
TEST 6: PostgreSQL FTS	tsvector query matched with rank score 25.5 on IOT Notes.pdf	PASSED (GIN FTS)
TEST 7: Lexical Comparison	Composite similarity 17%, 287 shared concepts, passage diffs	PASSED (100% Math)
TEST 8: Annotations CRUD	Created, retrieved, and deleted note in PostgreSQL	PASSED (CRUD)
TEST 9: Error Handling	Invalid document ID gracefully returned 404 JSON	PASSED (404 Handled)

6. Viva Voce & Technical Defense Questions

Q1: How does the system guarantee zero LLM hallucinations during document Q&A?

A1: Grounded RAG architecture. Document text is pre-chunked and embedded with text-embedding-004. Queries calculate cosine similarity directly in PostgreSQL. Only verified chunks exceeding threshold similarity are injected into the prompt with a strict constraint to answer solely from excerpts and cite [Source X].

Q2: Why use native PostgreSQL tsvector rather than external Elasticsearch?

A2: Simplifies the operational stack. Native GIN-indexed tsvectors yield sub-millisecond keyword retrieval with ts_headline snippet highlighting, eliminating cluster maintenance, separate synchronization pipelines, and memory overhead.

Q3: What role does SHA-256 play during document upload?

A3: Cryptographic collision-resistant duplicate detection. The server computes the SHA-256 hash from uploaded bytes. If identical bytes exist, the upload rejects redundant disk usage and points the user to the existing analysis.